Example 5.1

Construct a forward difference table for the following data

x	0	10	20	30
y	0	0.174	0.347	0.518

Solution:

The Forward difference table is given below

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	0			
		0.174		
10	0.174		-0.001	
		0.173		-0.001
20	0.347		-0.002	
		0.171		
30	0.518			

Example 5.2

Construct a forward difference table for $y = f(x) = x^3 + 2x + 1$ for $x = 1,2,3,4,5$

Solution:

$y = f(x) = x^3 + 2x + 1$ for $x = 1,2,3,4,5$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	4				
		9			
2	13		12		
		21		6	
3	34		18		0
		39		6	
4	73		24		
		63			
5	136				

Example 5.3

By constructing a difference table and using the second order differences as constant, find the sixth term of the series 8,12,19,29,42…

Solution:

Let k be the sixth term of the series in the difference table

First we find the forward differences.

x	y	Δ	Δ^2
1	8		
		4	
2	12		3
		7	
3	19		3
		10	
4	29		3
		13	
5	42		$k-55$
		$k-42$	
6	k		

Given that the second differences are constant

$\therefore k - 55 = 3$

$k = 58$

$\therefore$ the sixth term of the series is 58

Example 5.4

Find (i) Δe^{ax} (ii) $\Delta^2 e^x$ (iii) $\Delta \log x$

Solution:

(i) $\Delta e^{ax} = e^{a(x+h)} - e^{x}$

$= e^{ax}.e^{h} - e^{ax} \qquad \left[\because a^{m+n} = a^{m}.a^{n}\right]$

$= e^{ax}\left[e^{h} - 1\right]$

(ii) $\Delta^{2} e^{x} = \Delta.\left[\Delta e^{x}\right]$

$= \Delta\left[e^{x+h} - e^{x}\right] = \Delta\left[e^{x}e^{h} - e^{x}\right]$

$= \Delta e^{x}\left[e^{h} - 1\right]$

$= \left(e^{h} - 1\right)\Delta e^{x}$

$= \left(e^{h} - 1\right).\left(e^{h} - 1\right).e^{x}$

$= \left(e^{h} - 1\right)^{2}.e^{x}$

(iii) $\Delta \log x = \log(x+h) - \log x$

$= \log\dfrac{x+h}{x}$

$= \log\left(\dfrac{x}{x} + \dfrac{h}{x}\right)$

$= \log\left(1 + \dfrac{h}{x}\right)$